

UPC WORKSHOP III

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Modeling Part

1. Select the problem

Two types of problems:

- Physics process (relatively) deterministic, more complex in details
- Physics process determined (partially) by yourself; less complex in details

2. Modeling procedure

Rough Modeling

- Neglect details and assume simplest cases
- Mostly in the range of classical mechanics
- Calculate a rough answer to the problem

Example:

Problem A: Space Diving

Consider a skydiver, carried by a rocket vertically up to an altitude, perhaps above the Earth's atmosphere, who then exits the rocket in a space suit with a parachute, to descend back down to the Earth's surface. Analyze the challenges and dangers in such a descent, assuming that the total mass of the skydiver, space suit, and parachute is 190 kg. What would be the maximum altitude from which a person could successfully descend to the surface?

Assume that only gravity and air resistance exists, only consider the landing speed of the skydiver

Problem B: FIFA Penalty Kicks

In the game of FIFA football/soccer, when a penalty kick is taken, one player takes a single shot at the goal, which is only defended by the opposing team's goalkeeper. The kick is taken with the ball at the penalty mark, a point centered 11 meters from the goal, which is a rectangle 7.32 meters wide and 2.44 meters tall. To avoid the goalkeeper, the shooting player often aims for an upper corner of the goal. What initial ball velocities and spins will result in a successful shot to an upper corner from the penalty mark? What initial ball velocity and spin should the shooting player attempt to create to have the best chance of avoiding the goalkeeper and making the goal?

Assume that ball's trajectory is a standard parabola and the goalkeeper only moves up, left, down straightly

Purpose:

- estimate the order of magnitude of the answer
- Discover where to add details to your model
- Get yourselves into the groove!!!

Try to finish in 2 hours

Barely need professional paper; basic textbooks are enough.

Detailed Modeling

Find your neglected details and try to refine them

The density of air (ρ) depends on temperature (T) and varies with height (h). Based on the force balance in the atmosphere and the temperature variation with altitude, we can calculate the change in air density (ρ) with height (h) in the ranges of 0-160km:

$$\rho = \rho_0 \left(1 - \frac{k_1 h}{T_0} \right)^{\frac{\mu g}{k_1 R} - 1}, \quad 0 \leq h < 10km, \quad (10)$$

where $\rho_0 = 1.225kg/m^3$, $k_1 = 6.5 \times 10^{-3}K/m$, $T_0 = 298K$,

$$\rho = \rho_1 e^{-\frac{\mu g}{RT_1}(h-h_1)}, \quad 10km \leq h < 20km, \quad (11)$$

where $\rho_1 = 0.4334kg/m^3$, $T_1 = 233K$, $h_1 = 10km$,

$$\rho = \rho_2 \left(1 - \frac{k_3(h-h_2)}{T_2} \right)^{\frac{\mu g}{k_3 R} - 1}, \quad 20km \leq h < 50km, \quad (12)$$

where $\rho_2 = 0.1009kg/m^3$, $k_3 = -1.8 \times 10^{-3}K/m$, $T_2 = 233K$, $h_2 = 20km$,

$$\rho = \rho_3 \left(1 - \frac{k_4(h-h_3)}{T_3} \right)^{\frac{\mu g}{k_4 R} - 1}, \quad 50km \leq h < 160km, \quad (13)$$

$$\begin{cases} m \frac{dv_{\parallel}}{dt} = 2m\omega v_{\perp} - \frac{1}{2}CS\rho v_{\parallel}^2 \\ -m \frac{d^2 h}{dt^2} = mg_0 \left(\frac{R}{R+h} \right)^2 - \frac{1}{2}CS\rho \left(\frac{dh}{dt} \right)^2 \\ v_{\perp} = -\frac{dh}{dt} \end{cases}$$

Combine knowledge in other fields

Electrodynamics (i.e. electromagnetic oscillation causing heat);

Fluid mechanics (i.e. skydiver trapped in the storm);

Thermotics (i.e. Clapeyron equation)

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- Brainstorm yourselves and come up with all the factors that could influence your model
- Still need strong assumptions, but no worry

Carefully design your physical process

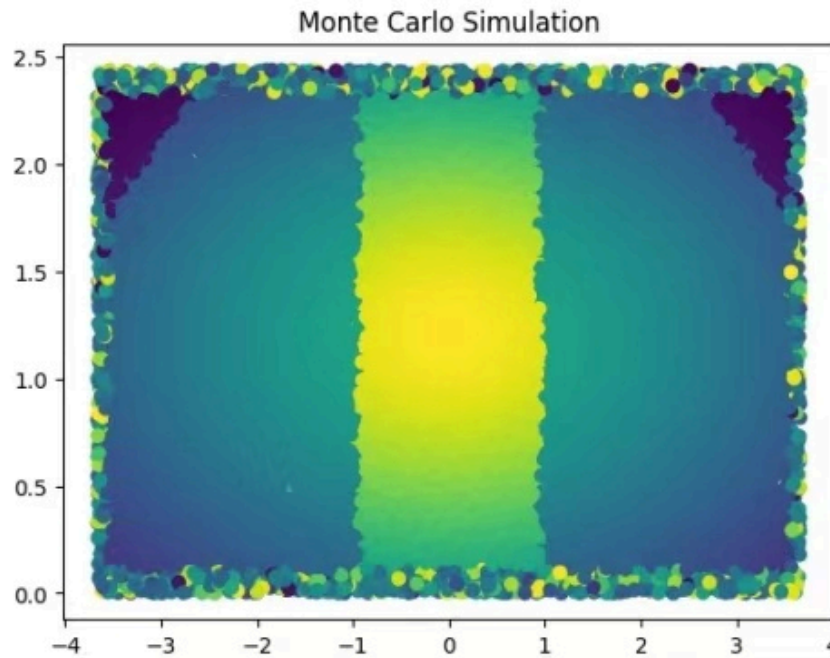


Figure 15: Distribution of success possibility

- Classified discussion
- Use advanced mathematics tools (Methods of mathematical physics, Differential geometry, PDE, etc.)
- Use coding and machine learning techniques (clustering; genetic algorithm, etc.)

General suggestions

- Pay high attention to the order of magnitude, neglect meaningless factors
- Refer to professional references (Google Scholar, SJTU Library, high-level textbooks, etc.)
- Give up factors that you're too unfamiliar and put them into assumption part

3. A few reminders

- Pay high attention to the timeline and check your progress
- Check if you made calculation mistakes carefully
- Practice on past year's problems (Problems are similar)

Enjoy your modeling time!